

Anirudh Ramkumar

Tamil Nadu, India | +91 99522 07837 | anirudhramkumar19@gmail.com | LinkedIn | GitHub

PROFILE

With a strong inclination towards silicon chip design, developed practical skills in design verification, microarchitecture design, and FPGA-based system design. Areas of interest include computer architecture, AI/ML hardware architectures, Embedded processing using FPGAs and SV/UVM-based verification. Passionate about lifelong learning, building in-depth understanding, and developing effective solutions in the evolving VLSI and AI landscape.

EDUCATION

PSG College of Technology, Coimbatore, India

Oct 2022 – May 2026

Bachelor of Engineering – Electronics and Communication, CGPA: 9.44/10

Relevant Coursework: Computer Architecture, FPGA-based System Design, Deep Learning, System-Level Verification, Analog VLSI Circuits, Computer Networks

EXPERIENCE

Design Verification Intern – Arm

Jan 2026 – Jun 2026

- Improved testcase portability across multicore configurations by adapting sequences for scalable execution on higher-core-count builds. Developed Power sequence for power-state transition verification, and debugged power-management behavior.
- Enabled multi-simulator UPF verification by debugging and modifying UPF constructs for compatibility across QuestaSim, Synopsys VCS, and Cadence Xcelium.

Project Intern – Aviat Networks

May 2025 – Dec 2025

- Designed and implemented port-based and MAC VLAN-based Layer-2 Ethernet switching on the Zynq-7000 APSoC.
- Identified RX descriptor recycling as the dominant throughput bottleneck in GEM DMA-based Ethernet reception under sustained ingress.
- Proposed and implemented a producer-consumer software pipeline using a bounded software ring buffer, improving sustained Ethernet throughput from 887 FPS to 16,000 FPS on a Cortex-A9 bare-metal platform.

PROJECTS

Low-Cost FPGA Implementation of HQC-128

Verilog, Vivado, Artix-7 FPGA

- Developed an area-optimized FPGA implementation of the HQC-128 post-quantum key encapsulation mechanism on Artix-7 FPGA.
- Redesigned intermediate storage and control architecture to improve BRAM inference and reduce reset-heavy LUT utilization. Achieved 10,472 LUT utilization with 36.10% lower LUT usage than existing balanced single-clock RTL implementations.
- Implemented and validated complete KeyGen, Encap, and Decap hardware flow with post-route Fmax of 112.233 MHz.

Single Stage RISC-V Core

Makerchip IDE

- Built a single-stage 32-bit RISC-V core using TL-Verilog with support for the RV32I instruction set.
- Verified functionality using assembly programs and waveform-based validation.

Wallace Tree Multiplier

Cadence Virtuoso

- Implemented a 4-bit Wallace tree multiplier and evaluated delay, power, and PDP across multiple adder architectures.
- Analyzed voltage-scaling tradeoffs for supply voltages ranging from 1.125V to 1.8V.

Design and Verification of 4-bit Adder/Subtractor

QuestaSim

- Implemented a structural Verilog design of a 4-bit adder/subtractor and verified functionality using a layered SystemVerilog testbench.
- Achieved 100% code coverage in QuestaSim.

Robotic Sorting Arm

Shakti SDK, FPGA Hackathon

- Developed a Shakti C-Class RISC-V processor-based robotic sorting arm on the Nexys Video FPGA platform.
- Implemented color-based object classification using the APDS9960 sensor.

Comparative Study of Current Mirrors

Cadence Virtuoso

- Implemented simple, source-degenerated, cascode, and Wilson current mirrors.
- Analyzed output resistance, I-V characteristics, and transistor aspect-ratio effects.

SKILLS

Verification: UVM(basics), power-aware verification

Protocols/Standards: AMBA, UPF(basics), P-channel, Q-channel

HDLs: Verilog, SystemVerilog, TL-Verilog(basics)

Programming Languages: Python, C, MATLAB, RISC-V & ARM assembly

Tools: Verdi, QuestaSim, Cadence Virtuoso, Xcelium, Synopsys VCS, NGSpice, xschem, Magic, Netgen, KLayout, Vivado, Vitis, Git, Vim, Makerchip

Hardware Platforms: Zynq-7000 APSoC, Zynq UltraScale+ MPSoC, Artix FPGAs, ESP8266, Arduino UNO, Raspberry Pi, 8051

ACTIVITIES AND ACHIEVEMENTS

- Finalist – SHAKTI FPGA Hackathon organized by IIT Madras and Shakra Innovations.
- GATE 2025 – AIR 2724.
- Completed Verification Part 1: Fundamentals of SystemVerilog offered by Namaste FPGA.
- Attended courses on System-Level Verification and Embedded Processing with FPGAs conducted by Qualcomm and NIELIT respectively.
- Third Place – Mathematics Olympiad 2023 organized by PSG College of Technology.
- NCC - 'A' certificate holder.
- Goju-Kai Karate - Brown III belt.